

$$E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{\sum_{i=1}^n E(X_i)}{n} = \mu$$

$\infty \quad E(X_i) = \mu$

$$\text{Var}\left[\frac{X_1 + X_2 + \dots + X_n}{n}\right] = \sum_{i=1}^n \text{Var}(X_i) / n^2 = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

$\rightarrow \{X_i\} \text{ are independent}$

$$Y = \sum_{i=1}^n X_i$$

$$\begin{aligned} \text{Var}(Y) &= E(Y^2) - (E(Y))^2 \\ &= E\left[\left(\sum_{i=1}^n X_i\right)^2\right] - \mu^2 \\ &= \frac{1}{n^2} \left(\sum_{i=1}^n E(X_i^2) + \sum_{i \neq j} 2E(X_i)E(X_j) \right) - \mu^2 \end{aligned}$$

$$\forall x, y \quad f_{XY}(x, y) = f_X(x) f_Y(y)$$

$$\int_{-\infty}^x \int_{-\infty}^y f_{XY}(x, y) dx dy = \int_{-\infty}^x \int_{-\infty}^y f_X(x) f_Y(y) dx dy$$

$$F_{XY}(x, y) = \int_{-\infty}^x f_X(x) dx \int_{-\infty}^y f_Y(y) dy$$

$$\rightarrow F_X(x) F_Y(y)$$